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Forecasting Time Series Data, STAT-GB 6018

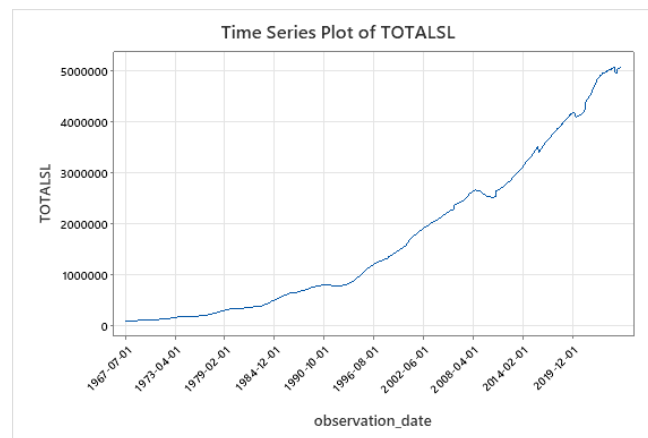
2 November, 2025

FORECASTING PROJECT 1

Source of Data:

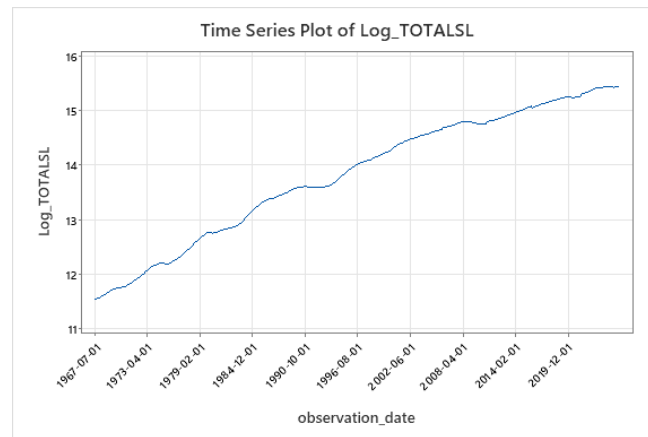
- Name: Total Consumer Credit Owned and Securitized (TOTALSL)
- Source: FRED, Federal Reserve Bank of St. Louis
- URL: <https://fred.stlouisfed.org/series/TOTALSL>
- Frequency: Monthly
- Observation period: 1967-07-01 to 2025-08-01

I first plotted the time series of TOTALSL to visually inspect its overall trend and patterns. The plot is shown as follows:



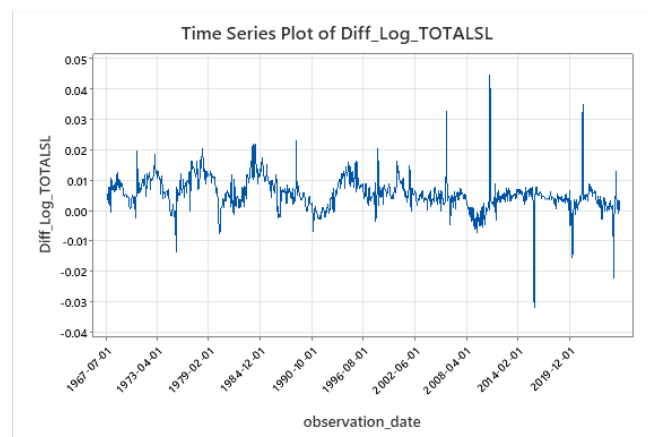
The plot clearly shows a strong upward trend over time, with no obvious seasonal fluctuations. Since the series appears to increase at an exponential level, I considered applying a **logarithmic transformation** to stabilize the variance.

Then I plotted the time series of Log TOTALSL to check its overall trend and patterns. The plot is shown as follows:

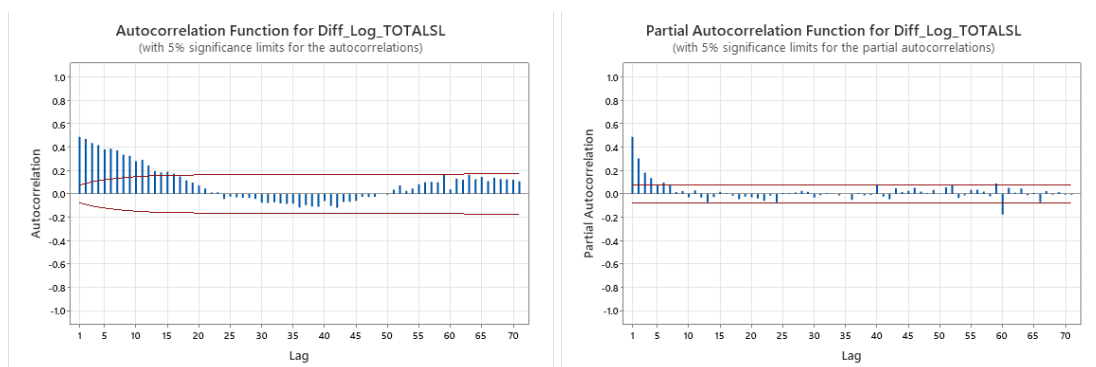


Now it shows that the series is increasing almost linearly, suggesting that **taking logs is necessary**.

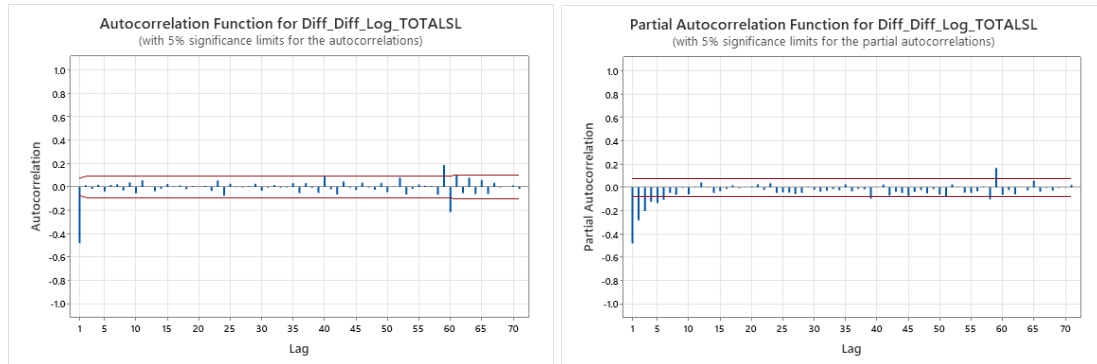
Since the series still exhibits a clear trend, which means it's not stationary, I applied a first difference to eliminate the trend. The time series plot of the first difference of Log TOTALSL is shown as follows:



Now it's stationary. I then examined the ACF and PACF of the first differencing of Log TOTALSL. The plots of ACF and PACF are shown as follows:



Both plots show dying down, suggesting that the first difference may be sufficient to achieve stationarity. To confirm this, a second differencing was applied to check for possible over-differencing. Then I examined the ACF and PACF of the second differencing of Log TOTALSL. The plots of ACF and PACF are shown as follows:



Both plots show strong negative autocorrelations at the first lag, indicating that it is **over-differencing**. Based on these results, it can be concluded that a **first difference ($d=1$)** is appropriate for modeling the series.

Then, I used **AICc** to select the proper *ARIMA* model, using $d = 1$ and $N=699$. I conducted forecasts with several candidate ***ARIMA(p,1,q)*** models and compared their *AICc* values by using the “Forecast with Best *ARIMA* Model” function. The model with the lowest *AICc* should be chosen as the best *ARIMA* model for the Log TOTAL series. The data are shown as follows (the first one includes the constant term while the second one doesn't):

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 1, q = 1*	2791.88	-5575.70	-5575.76	-5557.57
p = 2, q = 1	2791.94	-5573.79	-5573.88	-5551.14
p = 1, q = 2	2791.94	-5573.79	-5573.88	-5551.14
p = 3, q = 3	2794.43	-5572.66	-5572.87	-5536.49
p = 2, q = 2	2791.99	-5571.86	-5571.98	-5544.70
p = 3, q = 1	2791.94	-5571.75	-5571.87	-5544.59
p = 1, q = 3	2791.93	-5571.75	-5571.87	-5544.59
p = 1, q = 4	2792.08	-5570.01	-5570.17	-5538.34
p = 2, q = 5	2794.12	-5569.98	-5570.24	-5529.32
p = 4, q = 1	2792.05	-5569.94	-5570.10	-5538.27
p = 2, q = 3	2791.99	-5569.81	-5569.98	-5538.15
p = 3, q = 2	2791.99	-5569.81	-5569.98	-5538.15
p = 4, q = 5	2795.86	-5569.33	-5569.72	-5519.70
p = 1, q = 5	2792.18	-5568.15	-5568.36	-5531.98
p = 2, q = 4	2792.09	-5567.97	-5568.18	-5531.80
p = 3, q = 4	2793.11	-5567.96	-5568.22	-5527.30
p = 5, q = 1	2792.08	-5567.95	-5568.15	-5531.78
p = 4, q = 2	2792.07	-5567.92	-5568.13	-5531.76
p = 4, q = 3	2793.07	-5567.87	-5568.14	-5527.21
p = 5, q = 3	2793.11	-5565.89	-5566.21	-5520.74
p = 5, q = 2	2792.06	-5565.86	-5566.13	-5525.20
p = 3, q = 5	2793.09	-5565.86	-5566.18	-5520.71
p = 4, q = 4	2792.86	-5565.39	-5565.71	-5520.24
p = 5, q = 4	2793.13	-5563.87	-5564.26	-5514.24
p = 5, q = 0	2788.11	-5562.05	-5562.21	-5530.39
p = 4, q = 0	2786.24	-5560.35	-5560.47	-5533.19
p = 3, q = 0	2779.69	-5549.28	-5549.37	-5526.64
p = 2, q = 0	2767.69	-5527.33	-5527.39	-5509.20
p = 0, q = 5	2759.77	-5505.37	-5505.53	-5473.70
p = 0, q = 4	2755.00	-5497.88	-5498.00	-5470.72
p = 0, q = 3	2743.09	-5476.10	-5476.19	-5453.46
p = 1, q = 0	2733.96	-5461.88	-5461.91	-5448.27
p = 0, q = 2	2725.71	-5443.36	-5443.42	-5425.23
p = 0, q = 1	2695.87	-5385.71	-5385.75	-5372.11
p = 0, q = 0	2638.49	-5272.96	-5272.98	-5263.88

* Best model with minimum AICc.

Model Selection

Model (d = 1)	LogLikelihood	AICc	AIC	BIC
p = 1, q = 1*	2785.61	-5565.19	-5565.23	-5551.59
p = 2, q = 1	2785.61	-5563.16	-5563.22	-5545.03
p = 1, q = 2	2785.60	-5563.14	-5563.20	-5545.01
p = 3, q = 3	2787.84	-5561.52	-5561.68	-5529.85
p = 1, q = 3	2785.66	-5561.24	-5561.33	-5538.60
p = 2, q = 2	2785.64	-5561.20	-5561.29	-5538.55
p = 3, q = 1	2785.63	-5561.17	-5561.26	-5538.52
p = 2, q = 3	2785.71	-5559.29	-5559.41	-5532.13
p = 3, q = 2	2785.69	-5559.27	-5559.39	-5532.11
p = 4, q = 1	2785.69	-5559.26	-5559.38	-5532.10
p = 5, q = 4	2789.20	-5558.08	-5558.40	-5512.93
p = 4, q = 4	2788.15	-5558.05	-5558.31	-5517.39
p = 2, q = 4	2785.81	-5557.45	-5557.62	-5525.79
p = 4, q = 2	2785.74	-5557.32	-5557.48	-5525.65
p = 5, q = 1	2785.71	-5557.26	-5557.43	-5525.60
p = 1, q = 5	2785.70	-5557.24	-5557.40	-5525.57
p = 4, q = 3	2786.72	-5557.23	-5557.44	-5521.06
p = 3, q = 4	2785.87	-5555.54	-5555.75	-5519.38
p = 3, q = 5	2786.89	-5555.51	-5555.77	-5514.85
p = 5, q = 3	2786.87	-5555.48	-5555.75	-5514.83
p = 4, q = 5	2786.86	-5553.39	-5553.71	-5508.24
p = 5, q = 5	2787.62	-5552.85	-5553.23	-5503.22
p = 5, q = 0	2777.37	-5542.61	-5542.73	-5515.45
p = 4, q = 0	2773.46	-5536.83	-5536.91	-5514.18
p = 3, q = 0	2762.26	-5516.46	-5516.52	-5498.33
p = 2, q = 0	2740.91	-5475.78	-5475.81	-5462.17
p = 0, q = 5	2686.70	-5361.27	-5361.39	-5334.11
p = 1, q = 0	2677.97	-5351.92	-5351.94	-5342.85
p = 0, q = 4	2672.31	-5334.53	-5334.61	-5311.88
p = 0, q = 3	2643.06	-5278.06	-5278.12	-5259.93
p = 0, q = 2	2602.25	-5198.47	-5198.51	-5184.87
p = 0, q = 1	2532.62	-5061.21	-5061.23	-5052.14
p = 0, q = 0	2391.83	-4781.65	-4781.66	-4777.11

* Best model with minimum AICc.

Based on the data of the values of $AICc$ in different models, I identified that the best $ARIMA$ model for Log TOTALSL series is **$ARIMA(1,1,1)$ with constant term**, which generates the lowest $AICc$ of -5575.70.

The estimates of parameters for the selected model are shown as follows:

Final Estimates of Parameters

Type	Coef	SE Coef	T-Value	P-Value
AR 1	0.9452	0.0174	54.19	0.000
MA 1	0.7002	0.0378	18.55	0.000
Constant	0.000300	0.000050	5.96	0.000

Differencing: 1 Regular

Number of observations after differencing: 697

Thus, the estimated parameters are:

- $AR(1) = 0.9452$ ($p = 0.000$)
- $MA(1) = 0.7002$ ($p = 0.000$)
- $Constant = 0.000300$ ($p = 0.000$)

Based on the p-values, all of the estimated parameters are statistically significant at the 5% level. With these parameters, the fitted model is:

$$x_t = 0.9452x_{t-1} + \varepsilon_t - 0.7002\varepsilon_{t-1} + 0.000300$$

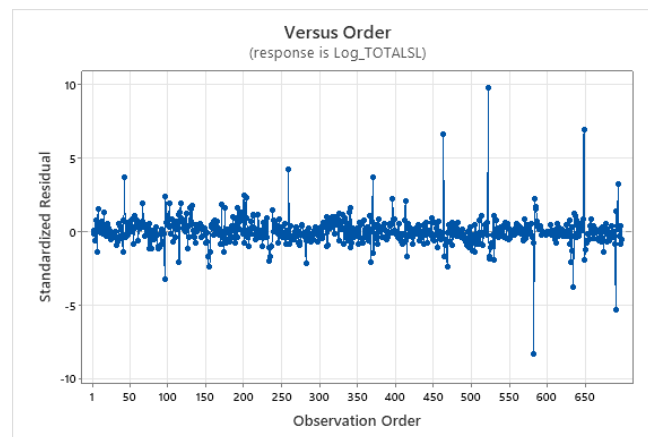
Then I examined the Ljung-Box statistics for lack of fit for the $ARIMA(1,1,1)$ model with the constant term, which is:

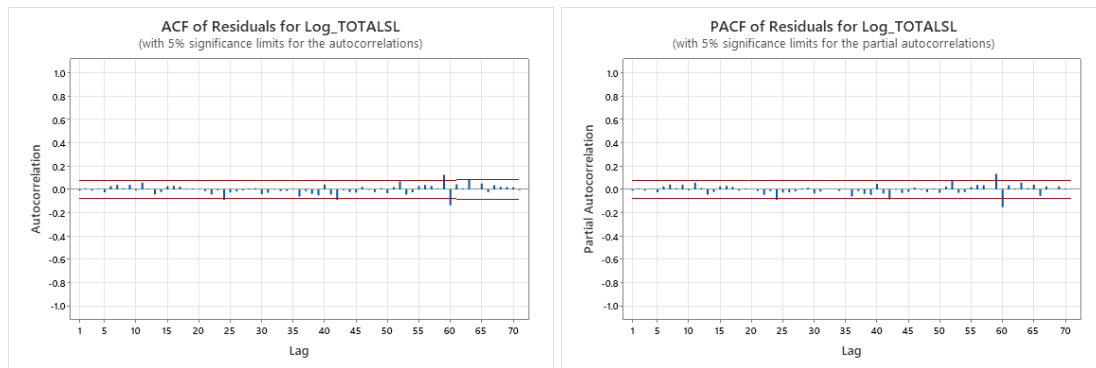
Modified Box-Pierce (Ljung-Box) Chi-Square Statistic

Lag	12	24	36	48
Chi-Square	6.09	16.70	22.34	35.99
DF	9	21	33	45
P-Value	0.731	0.729	0.920	0.829

According to the Ljung-Box test results in the table, all p-values (0.731, 0.729, 0.920, and 0.829) are much greater than 0.05 at every DF degrees. This means the Chi-Square statistics do not exceed the 95th percentile of the chi-square distribution with their respective degrees of freedom. Therefore, the model appears to be **adequate**.

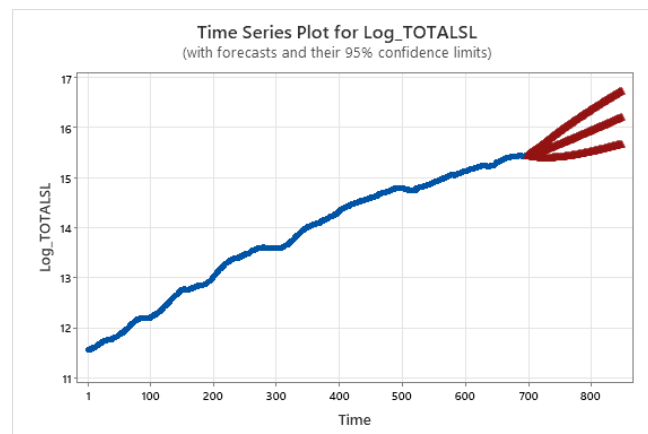
For diagnostic checking, I plotted the residuals as well as the ACF and PACF of the residuals from the fitted model, which are shown as follows:





The residuals versus order plot shows no visible pattern or trend, indicating that the residuals are randomly distributed over time and resemble white noise. Both the ACF and PACF plots show no significant spikes outside the confidence bounds, indicating that the residuals behave like an $ARIMA(0,0,0)$ process, which is white noise. These results collectively suggest that the residuals approximate white noise, confirming that the model has successfully captured the underlying structure of the data, and that there are **no noticeable inadequacies** remaining in the residuals.

After the diagnostic checking, I plotted the forecast of Log TOTALSL based on the $ARIMA(1,1,1)$ model that I got. The forecasts and 95% forecast intervals for lead times 1 to 150 are shown as follows:



The forecasts for Log TOTALSL appear **reasonable** because they follow the established upward trend in the historical data and show a gradual leveling off in the projected period. This pattern reflects a plausible and logical continuation of past observations, suggesting that the model is successfully capturing both the trend and the changing rate of increase in the series.

The 95% forecast intervals on the plot are **appropriately sized** and increase in width with the time, which is expected due to rising uncertainty further into the future. These intervals do not seem excessively wide, meaning the model provides a useful degree of precision while still acknowledging unavoidable forecasting uncertainty.

In summary, this report analyzes the Total Consumer Credit Owned and Securitized (TOTALSL) monthly time series from July 1967 to August 2025, confirming a strong upward trend with no notable seasonality. To stabilize the data, a logarithmic transformation was applied, producing a nearly linear series, which was then differenced to achieve stationarity. The *ARIMA*(1,1,1) model with a constant term was chosen using ACF, PACF, and *AICc*, and parameter significance and diagnostic checks confirmed model adequacy. Finally, forecasts and the 95% confidence intervals generated with this model appropriately capture the observed trend and feature expanding. This result shows a consistent and strong upward trend in total U.S. consumer credit over the past decades, reflecting continuous growth in borrowing activity.